13. <u>Solution 1.</u> (motivated by the def of limit).

$$\lim_{n\to+\infty}(2n)^{\frac{1}{n}} = \lim_{n\to+\infty} 2^{\frac{1}{n}} \cdot \lim_{n\to+\infty} n^{\frac{1}{n}}.$$

- $$2^{\frac{1}{n}} = (2\cdot 1\cdot\cdots\cdot 1)^{\frac{1}{n}} \underset{\substack{\uparrow \\ \text{mean inequality}}}{\le} \frac{2+1+\cdots+1}{n} = \frac{2+1\cdot(n-1)}{n} = \frac{n+1}{n} \xrightarrow{n\to+\infty} 1.$$

- prove $\lim_{n\to+\infty} n^{\frac{1}{n}} = 1$. Here is a standard trick:

Denote $a_n := n^{\frac{1}{n}} - 1$. ① $a_n \geq 1^{\frac{1}{n}} - 1 = 1 - 1 = 0$.

② $n^{\frac{1}{n}} = a_n + 1 \Rightarrow n = (a_n+1)^n = 1 + C_n^1 \cdot 1^{n-1} a_n + C_n^2 \cdot 1^{n-2} a_n^2 + \dots + a_n^n$

(Since $a_n \geq 0$, only reserve this term) $\geq 0 + 0 + \frac{n(n-1)}{2} \cdot a_n^2 + 0 + \dots + 0$ (★).

i.e. $n \geq \frac{n(n-1)}{2} a_n^2 \Rightarrow 0 \le a_n^2 \le \frac{2}{n-1} \xrightarrow{n\to+\infty} 0.$

$$\Rightarrow \lim_{n\to+\infty} a_n = 0 \iff \lim_{n\to+\infty} n^{\frac{1}{n}} = 1.$$

(If use "$\varepsilon$-N", (★) is the key, actually we use (★) to estimate $|n^{\frac{1}{n}} - 1|$.) □

<u>Solution 2</u> (My favorite! By <u>Stolz's Theorem</u>).

Take the ln: $\lim_{n\to+\infty}(2n)^{\frac{1}{n}} = 1 \iff \lim_{n\to+\infty} \frac{\ln(2n)}{n} = 0.$

By Stolz's Th: $$\lim_{n\to+\infty} \frac{\ln(2n)}{n} = \lim_{n\to+\infty} \frac{\ln(2n+2) - \ln(2n)}{(n+1) - n}$$

$$= \lim_{n\to+\infty} \ln\left(\frac{2n+2}{2n}\right) = \lim_{n\to+\infty} \ln\left(\frac{n+1}{n}\right) = \ln(1) = 0.$$ □.

[Highly recommand. Stolz's Th is the "sequence version" L'Hopital's Rule.

It works very well for the fraction style limit. ☺ ]

Scanned with CamScanner

14. <u>Solution 1.</u> (By Stolz's Th).

$$\lim_{n\to+\infty}\frac{n^2}{n!} \underset{\text{simplify}}{=} \lim_{n\to+\infty}\frac{n}{(n-1)!} \underset{\text{Stolz}}{=} \lim_{n\to+\infty}\frac{(n+1)-n}{n!-(n-1)!} \underset{\text{simplify}}{=} \lim_{n\to+\infty}\frac{1}{(n-1)!\cdot(n-1)} \underset{\geq \frac{1}{n-1}\xrightarrow{n\to+\infty}0}{=} 0. \quad \square.$$

<u>Solution2.</u> (For the form "$n^n$" "$n!$" "$n^\alpha$ ($\alpha>0$, is a power)". ln trick).

$$\lim_{n\to+\infty}\ln\left(\frac{n^2}{n!}\right) = \lim_{n\to+\infty}\ln\left(\frac{n}{(n-1)!}\right) = \lim_{n\to+\infty} \underbrace{\ln n}_{\text{group}} - \ln 1 - \ln 2 - \cdots - \underbrace{\ln(n-1)}_{\text{group}}.$$

$$= \lim_{n\to+\infty} - \ln 1 - \ln 2 - \cdots - \underbrace{\ln(n-2)} + \underbrace{\ln\left(\frac{n}{n-1}\right)}$$

Since $-\ln(n-2) \xrightarrow{n\to+\infty} -\infty$ and $\ln\left(\frac{n}{n-1}\right) \xrightarrow{n\to+\infty} \ln 1 = 0.$

$= -\infty.$

i.e. $\lim_{n\to+\infty}\frac{n^2}{n!} = 0.$ $\square.$

♡ <u>Solution3.</u> $0 \leq \frac{n^2}{n!} = \frac{n}{(n-1)!} \underset{\text{when } n\geq 3}{\leq} \frac{n}{(n-1)\cdot(n-2)} \xrightarrow{n\to+\infty} 0.$ So $\lim_{n\to+\infty}\frac{n^2}{n!} = 0.$

(I prefer solution3. Since if we're asked to use "$\varepsilon$-N" def to prove, solution3 still works. But solution 1,2 are very important tricks).

Scanned with CamScanner

15. <u>Solution 1.</u> (Stolz's Th).

Step 1. (prove the limit exist (and finite)).

Denote $a_n := \frac{2^n}{n!} \geq 0$. $\quad \frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{2^n} \cdot \frac{n!}{(n+1)!} = \frac{2}{n+1} \underset{\substack{\uparrow \\ \text{when } n \geq 1}}{\leq} 1.$

So $a_n$ decreasing. has the lower bound 0. $\Rightarrow$ limit exist and finite !.

Step 2. (use Stolz's Th to compute).

$$\boxed{\lim_{n\to+\infty} \frac{2^n}{n!}} \underset{\substack{\uparrow \\ \text{Stolz}}}{=} \lim_{n\to+\infty} \frac{2^{n+1}-2^n}{(n+1)!-n!} \underset{\substack{\uparrow \\ \text{simplify}}}{=} \lim_{n\to+\infty} \frac{(2-1)\cdot 2^n}{(n+1-1)\cdot n!} = \underbrace{\lim_{n\to+\infty} \frac{2}{n}}_{\substack{\| \\ 0.}} \cdot \boxed{\lim_{n\to+\infty} \frac{2^n}{n!}}$$

Denote the $\boxed{\text{limit} \ldots} = a$. i.e. $a = 0 \cdot a$ and $a$ is finite. $\Rightarrow a = 0$. $\square$

<u>Solution 2.</u> (Follow the Hint.)

$$0 < \frac{2^n}{n!} = \frac{2 \cdot \cancel{2} \cdot 2 \cdot 2^{n-3}}{1 \cdot \cancel{2} \cdot 3 \cdot 4 \cdots n} = 2 \cdot 1 \cdot \frac{2}{3} \cdot \underbrace{\frac{2}{4} \cdot \cdots \cdot \frac{2}{n}} \leq 2 \cdot 1 \cdot \frac{2}{3} \cdot \underbrace{\frac{2}{3} \cdot \cdots \cdot \frac{2}{3}}$$

$$= 2 \cdot \left(\frac{2}{3}\right)^{n-2} \xrightarrow{n\to+\infty} 0 \quad \square$$

$\uparrow$ since $\frac{2}{3} < 1$.

( I prefer solution 2. Inequality is the most useful and beautiful thing ).

Scanned with CamScanner